

# Chapter 1

## Competing tendencies In Heavy Fermion Systems: Kondo screening vs. Mott localisation

### 1.1 Bilayer extended Hubbard model: Impurity model

[1] We approach the heavy-fermion problem by starting from a bilayer extended Hubbard model, consisting of two layers ( $f$  and  $c$ ). Towards studying this lattice model, we adopt a two-layer impurity problem that hosts a correlated impurity site in each layer ( $S_f$  and  $S_d$ ):

$$H_{\text{aux}} = H_{\text{lp}} + H_f + H_d + H_{fd} , \quad (1.1)$$

where  $H_{\text{lp}}$  is the Hamiltonian for the non-interacting itinerant electrons of either layer,

$$H_{\text{lp}} = -t \sum_{\sigma, \alpha} \sum_{\langle i, j \rangle} \left( c_{i, \sigma, \alpha}^\dagger c_{j, \sigma, \alpha} + \text{h.c.} \right) - \sum_{i, \sigma, \alpha} \mu_\alpha n_{i, \sigma, \alpha} , \quad (1.2)$$

such that  $\alpha$  sums over the two layers  $f$  and  $d$ .  $H_f$  and  $H_d$  describe the dynamics of the correlated impurity sites (and their local neighbourhood) in each layer:

$$\begin{aligned} H_f = & -\frac{U_f}{2} (n_{f, \uparrow} - n_{f, \downarrow})^2 + \sum_{Z \in \text{NN}} \left[ -V \sum_{\sigma} \left( c_{f, \sigma}^\dagger c_{Z, \sigma, f} + \text{h.c.} \right) + \frac{J}{4} \sum_{\alpha, \beta} \mathbf{S}_f \cdot \frac{1}{2} \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \alpha, f}^\dagger c_{Z, \beta, f} \right. \\ & \left. - \frac{W_f}{4} (n_{Z, \uparrow, f} - n_{Z, \downarrow, f})^2 \right] , \\ H_d = & -\eta_d n_d - \frac{U_d}{2} (n_{d, \uparrow} - n_{d, \downarrow})^2 + \sum_{Z \in \text{NN}} \left[ -V \sum_{\sigma} \left( c_{d, \sigma}^\dagger c_{Z, \sigma, d} + \text{h.c.} \right) + \frac{J}{4} \sum_{\alpha, \beta} \mathbf{S}_d \cdot \frac{1}{2} \boldsymbol{\sigma}_{\alpha\beta} c_{Z, \alpha, d}^\dagger c_{Z, \beta, d} \right] , \end{aligned} \quad (1.3)$$

where  $c_{v, \sigma, \alpha}^\dagger$  can refer to creation operator for either the  $f$ -layer. Finally,  $H_{fd}$  represents the inter-layer hybridisation:

$$H_{fd} = -V_\perp \sum_{\sigma} \left( c_{d, \sigma}^\dagger c_{f, \sigma} + \text{h.c.} \right) . \quad (1.4)$$

## 1.2 Tiling Into Bilayer Extended Hubbard Model

Tiling the impurity model leads to a bilayer extended Hubbard model on the 2D square lattice. In order to tile our auxiliary model to restore translation invariance, we identify the cluster which acts as the “unit” cell of translation. As discussed in the text surrounding eq. ??, a good choice for the unit cell is the impurity site and its surrounding correlated environment. Placing the impurity sites at  $\mathbf{r}_d$  in each layer, we get

$$H_{\text{clust}}(\mathbf{r}_d) = H_f(\mathbf{r}_d) + H_d(\mathbf{r}_d) + H_{fc}(\mathbf{r}_d) . \quad (2.1)$$

The details of tiling a cluster Hamiltonian were already shown in Sec. ??; the general idea is to use manybody translation operators (details in Sec. ??) to center the cluster Hamiltonian at various sites of the lattice, such that combining all such terms leads to a periodised Hamiltonian. we write down the final result here:

$$\begin{aligned} H_{\text{latt}} &= \sum_{\mathbf{r}} T^\dagger(\mathbf{r}) H_{\text{clust}}(\mathbf{r}_d) T(\mathbf{r}) \\ &= -\frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left( c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} \right) - \sum_i \left[ \eta_d n_{i,d} + \frac{U_d}{2} (n_{i,\uparrow,d} - n_{i,\downarrow,d})^2 \right] + \frac{2J}{Z} \sum_{\langle i,j \rangle} \mathbf{S}_{i,d} \cdot \mathbf{S}_{j,d} \\ &\quad - \frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left( f_{i,\sigma}^\dagger f_{j,\sigma} + \text{h.c.} \right) - \frac{U_f - W_f}{2} \sum_i (n_{i,\uparrow,f} - n_{i,\downarrow,f})^2 + \frac{2J}{Z} \sum_{\langle i,j \rangle} \mathbf{S}_{i,f} \cdot \mathbf{S}_{j,f} \\ &\quad - V_\perp \sum_{i,\sigma} (f_{i,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) \end{aligned} \quad (2.2)$$

Tiling the impurity model therefore leads to a bilayer extended Hubbard model, with effective couplings that depend on the auxiliary model couplings. The value of  $\eta_d$  defines the effective chemical potential of the  $d$ -layer; a non-zero value moves the layer away from half-filling. The  $f$ -layer is pinned at half-filling.

Following Sec. ??, the retarded  $k$ -space Greens function in either layer can be expressed in terms of auxiliary model Greens functions:

$$G(\mathbf{k}, \sigma, \alpha; \omega) = \sum_{\nu} C_{\nu, k\alpha} \mathcal{G}(\mathcal{T}_{\nu, \sigma, \alpha}; \omega - \mathcal{E}_{\nu}(\mathbf{k})) \quad (2.3)$$

where  $\alpha = f, d$  denotes the layer in which the Greens function is being calculated, the modes  $\nu$  indicate the operators which diagonalise the Hamiltonian  $H_{\text{lp}}$ , with energies  $\mathcal{E}_{\nu}(\mathbf{k})$ :

$$H_{\text{lp}} = -\frac{2V}{Z} \sum_{\langle i,j \rangle, \sigma} \left( c_{i,\sigma}^\dagger c_{j,\sigma} + \text{h.c.} + f_{i,\sigma}^\dagger f_{j,\sigma} + \text{h.c.} \right) - V_\perp \sum_{i,\sigma} (f_{i,\sigma}^\dagger c_{i,\sigma} + \text{h.c.}) - \sum_i \eta_d n_{i,d} , \quad (2.4)$$

and the  $\mathcal{T}$ -matrices are defined in eq. ??:

$$\mathcal{T}_{\nu, \sigma, \alpha} = \left[ \left( \sum_{\sigma'} c_{\alpha\sigma'}^\dagger + \text{h.c.} \right) + (S_\alpha^+ + \text{h.c.}) + (C_\alpha^+ + \text{h.c.}) \right] c_{\nu, \sigma} , \quad (2.5)$$

In order to calculate  $\mathcal{E}_v(\mathbf{k})$ , we introduce the  $k$ -space modes via a Fourier transform:

$$c_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} c_{\mathbf{k},\sigma}^\dagger, \quad f_{i,\sigma}^\dagger = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{-i\mathbf{k} \cdot \mathbf{r}_i} f_{\mathbf{k},\sigma}^\dagger. \quad (2.6)$$

In terms of these modes, the 1-particle Hamiltonian decouples in  $k$ -space:

$$H_{1p} = \sum_{\mathbf{k},\sigma} \left[ (\epsilon_{\mathbf{k}} - \eta_d)(n_{\mathbf{k},\sigma,d} + n_{\mathbf{k},\sigma,f}) - V_\perp (c_{\mathbf{k},\sigma}^\dagger f_{\mathbf{k},\sigma} + \text{h.c.}) \right]. \quad (2.7)$$

The diagonal modes are given by

$$v_{\mathbf{k},\sigma,\pm}^\dagger = C_{\pm,\mathbf{k}} c_{\mathbf{k},\sigma}^\dagger + \sqrt{1 - C_{\pm,\mathbf{k}}^2} f_{\mathbf{k},\sigma}^\dagger, \quad (2.8)$$

the coefficients obtained from diagonalising the matrix

$$H_{1p}(\mathbf{k}) = \begin{pmatrix} \epsilon_{\mathbf{k}} & -V_\perp \\ -V_\perp & \epsilon_{\mathbf{k}} - \eta_d \end{pmatrix}. \quad (2.9)$$

The energies are given by  $\mathcal{E}_\pm(\mathbf{k}) = \epsilon_{\mathbf{k}} - \eta_d/2 \pm \Delta$ , with  $\epsilon_{\mathbf{k}}$  being  $-\frac{4V}{Z}(\cos k_x + \cos k_y)$  for a 2D square lattice and  $\Delta = \sqrt{V_\perp^2 + \eta_d^2}/4$ . Substituting into the expression for the lattice Greens function, we get

$$\begin{aligned} G(\mathbf{k}, d; \omega) = & C_{+,\mathbf{k}} \mathcal{G}(\mathcal{T}_{\mathbf{k}+,d}, \mathcal{T}_{\mathbf{k},d}^\dagger; \omega - \mathcal{E}_+(\mathbf{k})) + C_{-,\mathbf{k}} \mathcal{G}(\mathcal{T}_{\mathbf{k}-,d}, \mathcal{T}_{\mathbf{k},d}^\dagger; \omega - \mathcal{E}_-(\mathbf{k})) \\ & + C_{+,\mathbf{k}}^* \mathcal{G}(\mathcal{T}_{\mathbf{k},d}, \mathcal{T}_{\mathbf{k}+,d}^\dagger; \omega - \mathcal{E}_+(\mathbf{k})) + C_{-,\mathbf{k}}^* \mathcal{G}(\mathcal{T}_{\mathbf{k},d}, \mathcal{T}_{\mathbf{k}-,d}^\dagger; \omega - \mathcal{E}_-(\mathbf{k})). \end{aligned} \quad (2.10)$$

Following eq. ??, we can similarly write down an expression the local lattice Greens function for the  $f$ - and  $d$ -electrons. For that, we define the local diagonal modes  $v_\pm$  obtained from the local Hamiltonian

$$H_{\text{loc}} = \begin{pmatrix} 0 & -V_\perp \\ -V_\perp & 0 - \eta_d \end{pmatrix}, \quad (2.11)$$

and their associated energies  $\varepsilon_\pm$ . Within the impurity model, these local bonding/antibonding modes can be constructed on the impurity sites ( $v_\pm$ ) or on the neighbouring bath sites, ( $v_\pm^Z$ ). The local Greens function can be expressed in terms of these modes:

$$\begin{aligned} G_{dd}(\omega) = & \frac{1}{2} \sum_{\nu} C_{\nu,d} [\mathcal{G}(v\sigma, d\sigma; \omega - \varepsilon_\nu) + \mathcal{G}(d\sigma, v\sigma; \omega - \varepsilon_\nu)] \\ & + \frac{1}{2} \sum_{\nu^Z} C_{\nu^Z,Z_d} [\mathcal{G}(\mathcal{T}_{Z_d,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z}) + \mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z})]. \end{aligned} \quad (2.12)$$

The presence of local hybridisation between the  $d$ - and  $f$ - orbitals makes the inter-orbital Greens function important as well. Following eq. ??, we have

$$\begin{aligned} G_{fd}(\omega) = & \frac{1}{2} \sum_{\nu} [C_{\nu,f} \mathcal{G}(v\sigma, d\sigma; \omega - \varepsilon_\nu) + C_{\nu,d} \mathcal{G}(f\sigma, v\sigma; \omega - \varepsilon_\nu)] \\ & + \frac{1}{2} \sum_{\nu^Z} [C_{\nu^Z,Z_f} \mathcal{G}(\mathcal{T}_{\nu^Z,\sigma}, \mathcal{T}_{Z_d,\sigma}; \omega - \varepsilon_{\nu^Z}) + C_{\nu^Z,Z_d} \mathcal{G}(\mathcal{T}_{Z_f,\sigma}, \mathcal{T}_{\nu^Z,\sigma}; \omega - \varepsilon_{\nu^Z})]. \end{aligned} \quad (2.13)$$

Using these, we can construct the hybridised Greens function associated with the states  $v$ :

$$G_{v+}(\omega) = C_{+,d}^2 G_{dd}(\omega) + C_{+,f}^2 G_{ff}(\omega) + C_{+,d} C_{+,f} [G_{df}(\omega) + G_{fd}(\omega)] , \quad (2.14)$$

### 1.3 Unitary RG Analysis of The Bilayer Lattice-Embedded ESIAM

In the limit of large  $U_\alpha$ , we carry out a Schrieffer-Wolff transformation and work with the following low-energy Hamiltonian:

$$H_{\text{aux}} = \sum_{\mathbf{k}, \sigma, \alpha} \epsilon_{\mathbf{k}, \alpha} n_{\mathbf{k}, \sigma, \alpha} + \sum_{\alpha} \sum_{Z \in \text{NN}} \left[ \frac{1}{4} J_{\alpha} \sum_{\sigma, \sigma'} \mathbf{S}_{\alpha} \cdot \frac{1}{2} \sigma_{\sigma \sigma'} c_{Z, \sigma, \alpha}^{\dagger} c_{Z, \sigma', \alpha} - \frac{W_{\alpha}}{4} (n_{Z, \uparrow, \alpha} - n_{Z, \downarrow, \alpha})^2 \right] + J_{\perp} \mathbf{S}_f \cdot \mathbf{S}_d . \quad (3.1)$$

As we will see, additional terms are generated within the Hamiltonian as we integrate out high-energy degrees of freedom in the conduction baths of the two impurity models. These terms connect the impurity of the  $d$ -( $f$ )-layer to the nearest-neighbour conduction bath sites of the  $f$ -( $d$ )-layer, It is therefore necessary to start with an enlarged impurity model Hamiltonian that contains these couplings as well:

$$H_{\text{aux}} = \sum_{\mathbf{k}, \sigma, \alpha} \epsilon_{\mathbf{k}, \alpha} n_{\mathbf{k}, \sigma, \alpha} + \sum_{\alpha} \sum_{Z \in \text{NN}} \left[ \frac{1}{4} \sum_{\sigma, \sigma'} (J_{\alpha} \mathbf{S}_{\alpha} + K_{\bar{\alpha}} \mathbf{S}_{\bar{\alpha}}) \cdot \frac{1}{2} \sigma_{\sigma \sigma'} c_{Z, \sigma, \alpha}^{\dagger} c_{Z, \sigma', \alpha} - \frac{W_{\alpha}}{4} (n_{Z, \uparrow, \alpha} - n_{Z, \downarrow, \alpha})^2 \right] + J \mathbf{S}_f \cdot \mathbf{S}_d . \quad (3.2)$$

Note the additional couplings  $K_{\bar{\alpha}}$ , where  $\bar{\alpha}$  stands for the layer opposite to  $\alpha$ .

The two Kondo couplings  $J$  and  $K$  and the bath interaction  $W$  acquire  $k$ -dependence:

$$\begin{aligned} J(\mathbf{k}, \mathbf{q}) &= \frac{J}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)] , \\ K(\mathbf{k}, \mathbf{q}) &= \frac{K}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x) + \cos(\mathbf{k}_y - \mathbf{q}_y)] , \\ W(\mathbf{k}, \mathbf{q}, \mathbf{k}', \mathbf{q}') &= \frac{W}{2} [\cos(\mathbf{k}_x - \mathbf{q}_x + \mathbf{k}'_x - \mathbf{q}'_x) + \cos(\mathbf{k}_y - \mathbf{q}_y + \mathbf{k}'_y - \mathbf{q}'_y)] . \end{aligned} \quad (3.3)$$

The particle-hole transformed state  $\bar{\mathbf{q}}$  associated with  $\mathbf{q}$  is defined as  $\bar{\mathbf{q}} = \mathbf{q} + (\pi, \pi)$  if  $\mathbf{q}$  is in the first or third quadrant and  $\mathbf{q} + (\pi, -\pi)$  is in the second or fourth quadrant.

In order to study the low-energy physics of the impurity model, we iteratively integrate out high-energy degrees of freedom using the unitary RG method. Let the momentum states being decoupled from the two layers be  $|\mathbf{q}, \sigma, f\rangle$  and  $|\mathbf{q}, \sigma, d\rangle$  if they are above the Fermi energy (unoccupied in the low-energy configuration) and  $|\bar{\mathbf{q}}, \sigma, f\rangle$  and  $|\bar{\mathbf{q}}, \sigma, d\rangle$  if they are below the Fermi energy (occupied in the low-energy configuration) For every shell of conduction bath states of width  $\Delta D$  integrated out, the Hamiltonian couplings renormalise by folding in the effects of virtual excitations to these states.

The full set of impurity-mediated scattering processes that lead to UV-IR mixing is:

**Particle sector excitation** These involve excited states above the Fermi energy:

$$\begin{aligned} P_{\text{dir}}(\alpha) &= \frac{1}{2} \sum_{k,q} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1, \sigma_2} c_{q, \sigma_1, \alpha}^\dagger c_{k, \sigma_2, \alpha} , \\ P_{\text{ind}}(\alpha) &= \frac{1}{2} \sum_{k,q} K_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1, \sigma_2} c_{q, \sigma_1, \bar{\alpha}}^\dagger c_{k, \sigma_2, \bar{\alpha}} . \end{aligned} \quad (3.4)$$

**Hole sector** These involve excited states below the Fermi energy:

$$\begin{aligned} H_{\text{dir}}(\alpha) &= \frac{1}{2} \sum_{k,q} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1, \sigma_2} c_{k, \sigma_1, \alpha}^\dagger c_{q, \sigma_2, \alpha} , \\ H_{\text{ind}}(\alpha) &= \frac{1}{2} \sum_{k,q} K_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1, \sigma_2} c_{k, \sigma_1, \bar{\alpha}}^\dagger c_{q, \sigma_2, \bar{\alpha}} . \end{aligned} \quad (3.5)$$

**Mixed sector** These involve simultaneous particle and hole excitations:

$$\begin{aligned} M_{\text{dir}}(\alpha) &= \frac{1}{2} \sum_{q, \bar{q}} J_\alpha \mathbf{S}_\alpha \cdot \sigma_{\sigma_1, \sigma_2} c_{q, \sigma_1, \alpha}^\dagger c_{\bar{q}, \sigma_2, \alpha} , \\ M_{\text{ind}}(\alpha) &= \frac{1}{2} \sum_{q, \bar{q}} K_{\bar{\alpha}} \mathbf{S}_{\bar{\alpha}} \cdot \sigma_{\sigma_1, \sigma_2} c_{q, \sigma_1, \bar{\alpha}}^\dagger c_{\bar{q}, \sigma_2, \bar{\alpha}} . \end{aligned} \quad (3.6)$$

### 1.3.1 Particle/hole scattering processes: direct with direct

We already have the renormalisation group equations for the couplings  $J_\alpha$  in the case of  $J = K = 0$ , arising from  $P_{\text{dir}}$  and  $H_{\text{dir}}$ :

$$\frac{\Delta J_f(\mathbf{k}_1, \mathbf{k}_2)}{\Delta D} = - \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_f(\mathbf{k}_1, \mathbf{q}) J_f(\mathbf{q}, \mathbf{k}_2) + 4J_f(\mathbf{q}, \bar{\mathbf{q}}) W_f(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] G_f^{(0)}(\mathbf{q}) , \quad (3.7)$$

where  $\mathbf{q}$  sums over the UV states being integrated out,  $\rho(\mathbf{q})$  is the density of states for the momentum states being integrated out and  $G_f^{(0)}$  is the propagator for the excited intermediate state (for  $J = 0$ ):

$$G_\alpha^J(\mathbf{q}) = \frac{1}{\omega - |\epsilon_\alpha(\mathbf{q})|/2 + J_\alpha(\mathbf{q})/4 + W_\alpha(\mathbf{q})/2} . \quad (3.8)$$

An identical equations holds for  $\Delta J_d$ , with all  $f$ -quantities replaced with  $d$ -quantities. Switching on  $J$  modifies the excitation energy in the propagator, while either of the  $K$ s do not enter the denominator because either the impurity or the conduction bath labels do not match the ones fluctuating in this process:

$$G_\alpha(\mathbf{q}) = \frac{1}{1/G_\alpha^J - J \vec{S}_f \cdot \vec{S}_d} . \quad (3.9)$$

This expression can be simplified by noting that the operator in the denominator has only two eigenvalues,  $-3/4$  in the singlet sector and  $1/4$  in the triplet sector. Defining the projectors  $\mathcal{P}_0$  and  $\mathcal{P}_1$  for the two sectors, we have

$$\vec{S}_f \cdot \vec{S}_d = \frac{-3}{4} \mathcal{P}_0 + \frac{1}{4} \mathcal{P}_1, \quad \mathcal{P}_0 + \mathcal{P}_1 = \mathbb{I}. \quad (3.10)$$

Using these relations, we can write

$$\begin{aligned} G_f(\mathbf{q}) &= \frac{1}{1/G_f^J + 3J/4} \mathcal{P}_0 + \frac{1}{1/G_f^J - J/4} \mathcal{P}_1 \\ &= \left( \frac{1/4}{1/G_f^J + 3J/4} + \frac{3/4}{1/G_f^J - J/4} \right) \mathbb{I} + \left( \frac{1}{1/G_f^J + 3J/4} - \frac{1}{1/G_f^J - J/4} \right) \vec{S}_f \cdot \vec{S}_d. \end{aligned} \quad (3.11)$$

Only the first operator renormalises the Kondo interaction:

$$\frac{\Delta J_\alpha}{\Delta D} = - \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + 4J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \left( \frac{1}{4} \frac{1}{1/G_\alpha^J(\mathbf{q}) + 3J/4} + \frac{3}{4} \frac{1}{1/G_\alpha^J(\mathbf{q}) - J/4} \right). \quad (3.12)$$

We now simplify the second term, in conjunction with the two kinds of processes on either side:  $J^2$  processes and  $JW$  processes. We first focus on the  $J^2$  processes.

### Effect of inter-layer term $\vec{S}_f \cdot \vec{S}_d$ on $J^2$ processes

Taking into account the other spin operators on either side of  $G_f(\mathbf{q})$  in the full scattering process, we have

$$\frac{1}{2} \sum_{a,b,c} \sum_{\alpha,\beta} \sum_{\gamma,\delta} S_f^a S_{\alpha,\beta}^a S_f^b S_d^b S_f^c S_{\gamma,\delta}^c \left[ \delta_{\beta,\gamma} c_{k,\alpha,f}^\dagger c_{q,\beta,f} c_{q,\gamma,f}^\dagger c_{k',\delta,f} + \delta_{\alpha,\delta} c_{\bar{q},\alpha,f}^\dagger c_{k',\beta,f} c_{k,\gamma,f}^\dagger c_{\bar{q},\delta,f} \right]. \quad (3.13)$$

Integrating out the UV states using  $n_{\bar{q}\beta} = 1 - n_{q\beta} = 1$ , we get

$$\frac{1}{2} \sum_{a,b,c} \sum_{\alpha,\beta,\gamma} S_f^a S_f^b S_d^b S_f^c \left[ \sigma_{\alpha,\beta}^a \sigma_{\beta,\gamma}^c c_{k,\alpha,f}^\dagger c_{k',\gamma,f} + \sigma_{\alpha,\beta}^a \sigma_{\gamma,\alpha}^c c_{k',\beta,f} c_{k,\gamma,f}^\dagger \right]. \quad (3.14)$$

Using  $\sum_\beta \sigma_{\alpha,\beta}^a \sigma_{\beta,\gamma}^c = (\sigma^a \sigma^c)_{\alpha,\gamma} = \delta_{ac} \delta_{\alpha\gamma} + i\epsilon_{acm} \sigma_{\alpha,\gamma}^m$  in the first term and  $\sum_\alpha \sigma_{\gamma,\alpha}^c \sigma_{\alpha,\beta}^a = (\sigma^c \sigma^a)_{\gamma,\beta} = \delta_{ca} \delta_{\gamma\beta} - i\epsilon_{acm} \sigma_{\gamma,\beta}^m$  in the second term, relabelling  $\beta \rightarrow \gamma$  and  $\gamma \rightarrow \alpha$  there and using  $c_{k',\gamma} c_{k,\alpha}^\dagger = \delta_{k,k'} \delta_{\alpha,\gamma} - c_{k,\alpha}^\dagger c_{k',\gamma}$ , we get

$$\frac{1}{2} \sum_{a,b,c} \sum_{\alpha,\gamma} S_f^a S_f^b S_d^b S_f^c \left[ \delta_{a,c} \delta_{k,k'} \delta_{\alpha,\gamma} + 2i\epsilon_{acm} \sigma_{\alpha,\gamma}^m c_{k,\alpha,f}^\dagger c_{k',\gamma,f} \right]. \quad (3.15)$$

We also use the spin-half identity

$$\begin{aligned}
 \sigma^a \sigma^b \sigma^c &= (\delta_{ab} + i\epsilon_{abp} \sigma^p) \sigma^c \\
 &= \delta_{ab} \sigma^c + i\epsilon_{abp} (\delta_{pc} + i\epsilon_{pcr} \sigma^r) \\
 &= \delta_{ab} \sigma^c - \delta_{ac} \sigma^b + \delta_{bc} \sigma^a + i\epsilon_{abc} ,
 \end{aligned} \tag{3.16}$$

which when contracted with  $\epsilon_{acm}$  gives

$$\begin{aligned}
 \sum_{a,c} \epsilon_{acm} \sigma^a \sigma^b \sigma^c &= \sum_{a,c} \epsilon_{acm} (\delta_{ab} \sigma^c + \delta_{bc} \sigma^a + i\epsilon_{abc}) \\
 &= \sum_{a,c} (\epsilon_{acm} \delta_{ab} \sigma^c + \epsilon_{cam} \delta_{ba} \sigma^c - i\epsilon_{acm} \epsilon_{acb}) \\
 &= -2i\delta_{bm} .
 \end{aligned} \tag{3.17}$$

For the first term of eq. 3.15, we get

$$\begin{aligned}
 \frac{1}{2} \sum_{a,b,c} \sum_{\alpha,\gamma} S_f^a S_f^b S_d^b S_f^c \delta_{a,c} \delta_{k,k'} \delta_{\alpha,\gamma} &= \delta_{k,k'} \sum_{a,b} S_f^a S_f^b S_f^a S_d^b \\
 &= \frac{1}{4} \delta_{k,k'} \sum_b (S_f^b - 3S_f^b + S_f^b) S_d^b \\
 &= -\frac{1}{4} \delta_{k,k'} \sum_b S_f^b S_d^b .
 \end{aligned} \tag{3.18}$$

For the second term, we obtain

$$\begin{aligned}
 \frac{1}{2} \sum_{a,b,c} \sum_{\alpha,\gamma} S_f^a S_f^b S_d^b S_f^c 2i\epsilon_{acm} \sigma_{\alpha,\gamma}^m c_{k,\alpha,f}^\dagger c_{k',\gamma,f} &= \frac{1}{4} \sum_b \sum_{\alpha,\gamma} \delta_{bm} S_d^b \sigma_{\alpha,\gamma}^m c_{k,\alpha,f}^\dagger c_{k',\gamma,f} \\
 &= \frac{1}{2} \sum_b S_d^b \frac{1}{2} \sum_{\alpha,\gamma} \sigma_{\alpha,\gamma}^b c_{k,\alpha,f}^\dagger c_{k',\gamma,f}
 \end{aligned} \tag{3.19}$$

Substituting these in the expression for the renormalisation gives

$$\begin{aligned}
 \frac{1}{2} \sum_{a,b,c} \sum_{\alpha,\gamma} S_f^a S_f^b S_d^b S_f^c &\left[ \delta_{a,c} \delta_{k,k'} \delta_{\alpha,\gamma} + 2i\epsilon_{acm} \sigma_{\alpha,\gamma}^m c_{k,\alpha,f}^\dagger c_{k',\gamma,f} \right] \\
 &= -\frac{1}{4} \mathbf{S}_f \cdot \mathbf{S}_d \delta_{k,k'} + \frac{1}{2} \sum_{\alpha,\gamma} \mathbf{S}_d \frac{1}{2} \sigma_{\alpha,\gamma} c_{k,\alpha,f}^\dagger c_{k',\gamma,f} .
 \end{aligned} \tag{3.20}$$

The first term leads to the renormalisation of the inter-layer hybridisation  $J_\perp$ :

$$\frac{\Delta J_\perp}{\Delta D} = -\frac{1}{4} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) \int_{IR} d\mathbf{k} \rho(\mathbf{k}) \sum_\alpha J_\alpha(\mathbf{k}, \mathbf{q})^2 \left( \frac{1}{1/G_\alpha^J + 3J/4} + \frac{1}{1/G_\alpha^J - J/4} \right) , \tag{3.21}$$

while the second term gives rise to the indirect impurity-bath Kondo couplings  $K_\alpha$  mentioned earlier above eq. 3.2:

$$\frac{\Delta K_\alpha}{\Delta D} = \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) J_{\tilde{\alpha}}(\mathbf{k}_1, \mathbf{q}) J_{\tilde{\alpha}}(\mathbf{q}, \mathbf{k}_2) \left( \frac{1}{1/G_{\tilde{\alpha}}^J + 3J/4} + \frac{1}{1/G_{\tilde{\alpha}}^J - J/4} \right) , \tag{3.22}$$

### Effect of inter-layer term $\vec{S}_f \cdot \vec{S}_d$ on $JW$ processes

We now turn to the effect of the  $\vec{S}_f \cdot \vec{S}_d$  in eq. 3.11 on processes that involve a Kondo scattering  $J$  and a bath scattering  $W$ . The non-vanishing processes are of the form

$$\begin{aligned} P_1 &= (S_f^+ \mathbf{S}_f \cdot \mathbf{S}_d + \mathbf{S}_f \cdot \mathbf{S}_d S_f^+) c_{k,\downarrow}^\dagger c_{k',\uparrow}, \\ P_2 &= P_1^\dagger, \\ P_z &= (S_f^z \mathbf{S}_f \cdot \mathbf{S}_d + \mathbf{S}_f \cdot \mathbf{S}_d S_f^z) \sum_{\sigma} \sigma c_{k,\sigma}^\dagger c_{k',\sigma}. \end{aligned} \quad (3.23)$$

In order to evaluate them, we note that for any  $a \in x, y, z$ , we have

$$S_1^a \mathbf{S}_1 \cdot \mathbf{S}_2 + \mathbf{S}_1 \cdot \mathbf{S}_2 S_1^a = \sum_b \{S_1^a, S_1^b\} S_2^b = \frac{1}{4} \sum_b 2\delta_{a,b} S_2^b = \frac{1}{2} S_2^a. \quad (3.24)$$

Using this, we obtain

$$\begin{aligned} P_1 &= \frac{1}{2} S_d^+ c_{k,\downarrow}^\dagger c_{k',\uparrow}, \\ P_2 &= \frac{1}{2} S_d^- c_{k,\uparrow}^\dagger c_{k',\downarrow}, \\ P_z &= \frac{1}{2} S_d^z \sum_{\sigma} \sigma c_{k,\sigma}^\dagger c_{k',\sigma}. \end{aligned} \quad (3.25)$$

Since the bath sites are in the  $f$ -layer, these terms again lead to the renormalisation of the indirect couplings  $K_\alpha$ . Combining with eq. 3.22, we get

$$\begin{aligned} \frac{\Delta K_\alpha}{\Delta D} &= \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_{\bar{\alpha}}(\mathbf{k}_1, \mathbf{q}) J_{\bar{\alpha}}(\mathbf{q}, \mathbf{k}_2) - 4J_{\bar{\alpha}}(\mathbf{q}, \bar{\mathbf{q}}) W_{\bar{\alpha}}(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \left( \frac{1}{1/G_{\bar{\alpha}}^J + 3J/4} \right. \\ &\quad \left. + \frac{1}{1/G_{\bar{\alpha}}^J - J/4} \right), \end{aligned} \quad (3.26)$$

### 1.3.2 Particle/hole scattering processes: indirect with indirect

The processes that apply to this class are very similar to those participating in the direct-direct class; the only difference is that the bath degrees of freedom have to be transferred from  $\alpha \rightarrow \bar{\alpha}$ , due to the inter-layer nature of this coupling. As a result, we will have straightforward counterparts to the renormalisation equations obtained in the previous section, eqs. 3.12, 3.21 and 3.26. The first equation is for the renormalisation of the indirect coupling through itself,

$$\begin{aligned} \frac{\Delta K_\alpha}{\Delta D} &= - \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [K_\alpha(\mathbf{k}_1, \mathbf{q}) K_\alpha(\mathbf{q}, \mathbf{k}_2) + 4K_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_{\bar{\alpha}}(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \left( \frac{1}{4} \frac{1}{1/G_{\bar{\alpha}}^K(\mathbf{q}) + 3J/4} \right. \\ &\quad \left. + \frac{3}{4} \frac{1}{1/G_{\bar{\alpha}}^K(\mathbf{q}) - J/4} \right). \end{aligned} \quad (3.27)$$



where the indirect Greens function  $G_{\alpha}^K$  is defined as

$$G_{\bar{\alpha}}^K(\mathbf{q}) = \frac{1}{\omega - |\epsilon_{\bar{\alpha}}(\mathbf{q})|/2 + K_{\alpha}(\mathbf{q})/4 + W_{\bar{\alpha}}(\mathbf{q})/2}, \quad (3.28)$$

and reflects the fact the excitations are happening on impurity and bath states in opposite layers. The second equation is for the effect of the inter-layer term through the denominator of such processes, leading to a renormalisation of the inter-layer itself:

$$\frac{\Delta J_{\perp}}{\Delta D} = -\frac{1}{4} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) \int_{IR} d\mathbf{k} \rho(\mathbf{k}) \sum_{\alpha} K_{\alpha}(\mathbf{k}, \mathbf{q})^2 \left( \frac{1}{1/G_{\bar{\alpha}}^K + 3J/4} + \frac{1}{1/G_{\bar{\alpha}}^K - J/4} \right). \quad (3.29)$$

The final equation is for the renormalisation of the direct Kondo coupling arising from indirect processes. Similar to how the direct Kondo coupling renormalised the indirect coupling, this occurs through two routes. The complete equation is

$$\frac{\Delta J_{\alpha}}{\Delta D} = \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [K_{\bar{\alpha}}(\mathbf{k}_1, \mathbf{q}) K_{\bar{\alpha}}(\mathbf{q}, \mathbf{k}_2) - 4K_{\bar{\alpha}}(\mathbf{q}, \bar{\mathbf{q}}) W_{\alpha}(\mathbf{k}_1, \mathbf{q}, \mathbf{q}, \mathbf{k}_2)] \left( \frac{1}{1/G_{\alpha}^K + 3J/4} + \frac{1}{1/G_{\alpha}^K - J/4} \right). \quad (3.30)$$

### 1.3.3 Particle/hole scattering processes: direct with indirect

These processes are composed of a direct Kondo excitation process followed by an indirect Kondo de-excitation process, or vice versa. They can be of the particle excitation or the hole excitation kind. Within the particle or hole sector, we again have two sets of processes depending on whether the initial process is direct or indirect:

- a.  $P_{\text{dir}}(d) G_d H_{\text{ind}}(f) + \text{h.c.}$
- b.  $P_{\text{dir}}(f) G_f H_{\text{ind}}(d) + \text{h.c.}$
- c.  $H_{\text{ind}}(f) G_d P_{\text{dir}}(d) + \text{h.c.}$
- d.  $H_{\text{ind}}(d) G_f P_{\text{dir}}(f) + \text{h.c.}$

As mentioned above,  $a(b)$  and  $c(d)$  are time-reversed partners of each other. We will now show that the contributions from these processes lead to a renormalisation of the inter-layer Kondo term. We suppress the Greens function since it is the same for all the terms:

$$G_{\alpha}^{JK} = \frac{1}{\omega - |\epsilon_{\alpha}(\mathbf{q})|/2 + J_{\alpha}/4 + K_{\bar{\alpha}}/4 + W_{\alpha}(\mathbf{q})/2 - J_{\perp}/4}, \quad (3.31)$$

containing antiferromagnetic contributions from  $J_\alpha$  and  $K_\alpha$  and ferromagnetic contribution in  $J_\perp$ . The contribution from the pair is of the form

$$\begin{aligned} & \sum_{\alpha} [P_{\text{dir}}(\alpha)H_{\text{ind}}(\bar{\alpha}) + H_{\text{ind}}(\bar{\alpha})P_{\text{dir}}(\alpha)] \\ &= \sum_{\alpha,a,b,\beta,\gamma,\delta} \frac{1}{4} J_\alpha(q, k') K_{\bar{\alpha}}(k, q) \left[ \mathbf{S}_\alpha^a \sigma_{\beta,\gamma}^a c_{k',\gamma,\alpha} \mathbf{S}_{\bar{\alpha}}^b \sigma_{\delta,\beta}^b c_{k,\delta,\alpha}^\dagger + \mathbf{S}_{\bar{\alpha}}^b \sigma_{\delta,\beta}^b c_{k,\delta,\alpha}^\dagger \mathbf{S}_\alpha^a \sigma_{\beta,\gamma}^a c_{k',\gamma,\alpha} \right] \end{aligned} \quad (3.32)$$

The sum over  $\beta$  can be carried out:

$$\begin{aligned} & \sum_{\alpha,a,b,\beta,\gamma,\delta} \frac{1}{4} J_\alpha(q, k') K_{\bar{\alpha}}(k, q) \left[ \mathbf{S}_\alpha^a \sigma_{\beta,\gamma}^a c_{k',\gamma,\alpha} \mathbf{S}_{\bar{\alpha}}^b \sigma_{\delta,\beta}^b c_{k,\delta,\alpha}^\dagger + \mathbf{S}_{\bar{\alpha}}^b \sigma_{\delta,\beta}^b c_{k,\delta,\alpha}^\dagger \mathbf{S}_\alpha^a \sigma_{\beta,\gamma}^a c_{k',\gamma,\alpha} \right] \\ &= \sum_{\alpha,a,b,\gamma,\delta} \frac{1}{4} J_\alpha(q, k') K_{\bar{\alpha}}(k, q) (\sigma^b \sigma^a)_{\delta,\gamma} \mathbf{S}_\alpha^a \mathbf{S}_{\bar{\alpha}}^b \left[ c_{k',\gamma,\alpha} c_{k,\delta,\alpha}^\dagger + c_{k,\delta,\alpha}^\dagger c_{k',\gamma,\alpha} \right] \\ &= \sum_{\alpha,a,b,\gamma,\delta} \frac{1}{4} J_\alpha(q, k') K_{\bar{\alpha}}(k, q) (\sigma^b \sigma^a)_{\delta,\gamma} \mathbf{S}_\alpha^a \mathbf{S}_{\bar{\alpha}}^b \delta_{k,k'} \delta_{\delta,\gamma} \\ &= \sum_{\alpha,a,b} \frac{1}{4} J_\alpha(q, k') K_{\bar{\alpha}}(k, q) \text{Trace}(\sigma^b \sigma^a) \mathbf{S}_\alpha^a \mathbf{S}_{\bar{\alpha}}^b \delta_{k,k'} . \end{aligned} \quad (3.33)$$

Adding the contribution from the hermitian conjugate and restoring the Greens function, we get the total contribution as

$$\sum_{\alpha} [P_{\text{dir}}(\alpha)H_{\text{ind}}(\bar{\alpha}) + H_{\text{ind}}(\bar{\alpha})P_{\text{dir}}(\alpha)] + \text{h.c.} = \frac{1}{2} \mathbf{S}_d \cdot \mathbf{S}_f \sum_{\alpha,q,k} \frac{J_\alpha(q, k) K_{\bar{\alpha}}(k, q)}{1/G_\alpha^{JK} - J_\perp/4} . \quad (3.34)$$

The renormalisation of the inter-layer coupling arising from this process is

$$\frac{\Delta J_\perp}{\Delta D} = \int_{UV} d\mathbf{q} \rho(\mathbf{q}) \int_{IR} d\mathbf{k} \rho(\mathbf{k}) \sum_{\alpha,q,k} \frac{J_\alpha(q, k) K_{\bar{\alpha}}(k, q)}{1/G_\alpha^{JK} - J_\perp/4} . \quad (3.35)$$

### 1.3.4 Mixed processes

We now turn to the renormalisation arising from particle-hole mixed processes (designated as  $M_\alpha$  above). There are four types of process:

$$\begin{aligned} P_1 &: \sum_{\alpha} M_{\text{dir}}(\alpha) G M_{\text{dir}}(\alpha) \\ P_2 &: \sum_{\alpha} M_{\text{ind}}(\alpha) G M_{\text{ind}}(\alpha) \\ P_3 &: \sum_{\alpha} [M_{\text{dir}}(\alpha) G M_{\text{ind}}(\bar{\alpha}) + \text{h.c.}] \\ P_4 &: \sum_{\alpha} [M_{\text{dir}}(\alpha) G M_{\text{ind}}(\alpha) + \text{h.c.}] \\ P_5 &: \sum_{\alpha} [M_{\text{dir}}(\alpha) G M_{\text{dir}}(\bar{\alpha}) + M_{\text{ind}}(\alpha) G M_{\text{ind}}(\bar{\alpha})] . \end{aligned} \quad (3.36)$$

We first note that  $P_1$  and  $P_2$  are equivalent to the ones considered in the direct-direct and indirect-indirect subsections, specifically the renormalisation of  $J_\perp$  in eq. 3.21 and its indirect counterpart, the only difference being that  $k$  will now range over the UV subspace instead of IR:

$$\begin{aligned} \frac{\Delta J_\perp}{\Delta D} = & -\frac{1}{4} \int_{UV} d\mathbf{q} \rho(\mathbf{q}') \int_{UV} d\mathbf{q}' \rho(\mathbf{q}') \sum_{\alpha} \left[ J_{\alpha}(\mathbf{q}', \mathbf{q})^2 \left( \frac{1}{1/G_{\alpha}^J + 3J/4} + \frac{1}{1/G_{\alpha}^J - J/4} \right) \right. \\ & \left. + K_{\alpha}(\mathbf{k}, \mathbf{q})^2 \left( \frac{1}{1/G_{\alpha}^K + 3J/4} + \frac{1}{1/G_{\alpha}^K - J/4} \right) \right]. \end{aligned} \quad (3.37)$$

The other equations in the same subsection – the ones pertaining to the renormalisation of  $J_{\alpha}$  or  $K_{\alpha}$  are not relevant here because they require the participation of IR bath operators. Similarly,  $P_3$  and  $P_4$  are equivalent to the ones considered in the previous subsection (eq. 3.35), the only difference being that the dummy variable  $k$  in the previous subsection would now have to range only over the UV subspace:

$$\frac{\Delta J_\perp}{\Delta D} = \int_{UV} d\mathbf{q} \rho(\mathbf{q}) \int_{UV} d\mathbf{q}' \rho(\mathbf{q}') \sum_{\alpha} \frac{J_{\alpha}(q, q') K_{\alpha}(q', q)}{1/G_{\alpha}^{JK} - J_{\perp}/4}. \quad (3.38)$$

We finally evaluate the contribution from processes  $P_4$  and  $P_5$ . In order to capture coherent scattering processes between the two layers arising from  $P_4$ , we project onto the following rotated basis of excited states:

$$\begin{aligned} |H\rangle_{\pm} &= \frac{1}{\sqrt{2}} \left( |H(\mathbf{q})\rangle_f \pm |H(\mathbf{q})\rangle_d \right), \\ \mathcal{P}(\mathbf{q})_H &= |H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_-, \end{aligned} \quad (3.39)$$

where  $|H(\mathbf{q})\rangle_{\alpha}$  is the state obtained upon exciting both states in the layer  $\alpha$ :  $|H(q)\rangle_{\alpha} = c_{\mathbf{q},\alpha}^{\dagger} c_{\bar{\mathbf{q}},\alpha} |L\rangle_f |L\rangle_d$ , and  $\mathcal{P}(\mathbf{q})_H$  projects onto this rotated excited basis. We focus on the spin-flip component  $JS_f^+ S_d^-$  of the Hamiltonian. There are two kinds of processes that renormalise this coupling. We first consider one that involves a spin-flip of the  $d$ -layer followed by a spin-flip of the  $f$ -layer:

$$\Delta H = \frac{1}{N^2} \sum_{\mathbf{q} \in UV} \langle L | \frac{1}{2} J_f(\bar{\mathbf{q}}, \mathbf{q}) S_f^+ c_{\mathbf{q}-\downarrow, f}^{\dagger} c_{\mathbf{q}+\uparrow, f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H \frac{1}{2} J_d(\mathbf{q}, \bar{\mathbf{q}}) S_d^- c_{\mathbf{q}\uparrow, d}^{\dagger} c_{\bar{\mathbf{q}}\downarrow, d} | L \rangle, \quad (3.40)$$

where  $|L\rangle = |L\rangle_f |L\rangle_d$  is the initial configuration for both layers.  $G$  is the propagator for the excited state:

$$G = \frac{1}{\omega - H_D}, \quad (3.41)$$

where  $H_D$  is the diagonal part of the Hamiltonian corresponding to the excited(intermediate) state. The diagonal part consists of the kinetic energy and the Ising part of the Kondo interaction:

$$H_D = \sum_{\mathbf{q}, \sigma \in \mathbf{q}_{\pm}, \alpha} \epsilon_{\alpha}(\mathbf{q}) \tau_{\mathbf{q}, \sigma} + \frac{1}{2} \sum_{\alpha} J_{\alpha} S_{\alpha}^z \sum_{\mathbf{q}, \sigma} \sigma n_{\mathbf{q}, \sigma, \alpha} + J_{\perp} \vec{S}_f \cdot \vec{S}_d. \quad (3.42)$$

Calculating the matrix elements of the propagator in the rotated basis gives

$$\begin{aligned} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H = & \frac{1}{2} \left( \frac{1}{1/G_{ff}^J - J_{\perp}/4} + \frac{1}{1/G_{dd}^J - J_{\perp}/4} \right) (|H\rangle_+ \langle H|_+ + |H\rangle_- \langle H|_-) \\ & + \frac{1}{2} \left( \frac{1}{1/G_{ff}^J - J_{\perp}/4} - \frac{1}{1/G_{dd}^J - J_{\perp}/4} \right) (|H\rangle_+ \langle H|_- + |H\rangle_- \langle H|_+), \end{aligned} \quad (3.43)$$

where

$$G_{\alpha\alpha}^J(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2} [\epsilon_{\alpha}(\mathbf{q}) - \epsilon_{\alpha}(\bar{\mathbf{q}})] + \frac{1}{2} J_{\alpha}(\mathbf{q}) + W_{\alpha}(\mathbf{q})}. \quad (3.44)$$

where  $\bar{\epsilon}_{\alpha}(\mathbf{q}) = \epsilon_f(\mathbf{q}) - \epsilon_f(\bar{\mathbf{q}})$  is the total excitation energy for the pair of states  $\mathbf{q}, \bar{\mathbf{q}}$ .

In eq. 3.40, since the first process is an excitation into a  $d$ -state and the second process is a de-excitation from an  $f$ -state, the expression can be simplified into

$$\begin{aligned} \langle L | S_f^{\dagger} c_{\mathbf{q}-\downarrow, f}^{\dagger} c_{\mathbf{q}+\uparrow, f} \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H S_d^{-} c_{\mathbf{q}+\uparrow, d}^{\dagger} c_{\mathbf{q}-\downarrow, d} | L \rangle &= S_f^{\dagger} S_d^{-} \langle H_f | \mathcal{P}(\mathbf{q})_H G \mathcal{P}(\mathbf{q})_H | H_d \rangle \\ &= \frac{1}{2} \left( \frac{1}{1/G_{ff}^J - J_{\perp}/4} + \frac{1}{1/G_{dd}^J - J_{\perp}/4} \right) S_f^{\dagger} S_d^{-}. \end{aligned} \quad (3.45)$$

The net Hamiltonian renormalisation is finally a sum over the contribution from each mode:

$$\Delta H = \frac{1}{8N^2} \sum_{\mathbf{q} \in UV} J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \left( \frac{1}{1/G_{ff}^J - J_{\perp}/4} + \frac{1}{1/G_{dd}^J - J_{\perp}/4} \right) S_f^{\dagger} S_d^{-}. \quad (3.46)$$

Converting the sums to integrals (assuming a density of states  $\rho(\mathbf{q})$ ) gives the final expression

$$\frac{\Delta H}{\Delta D} = \frac{1}{8N} S_f^{\dagger} S_d^{-} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \left( \frac{1}{1/G_{ff}^J(q) - J_{\perp}/4} + \frac{1}{1/G_{dd}^J(q) - J_{\perp}/4} \right). \quad (3.47)$$

The time-reversed partner of this process can be obtained simply by exchanging the  $d$ -interaction with the  $f$ -interaction. Since the renormalisation itself is symmetric, the contribution is equal to what we obtained here. The total renormalisation in the coupling for the term  $S_f^{\dagger} S_d^{-}$  is therefore

$$\frac{\Delta J_{\perp}}{\Delta D} = \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \left( \frac{1}{1/G_{ff}^J(q) - J_{\perp}/4} + \frac{1}{1/G_{dd}^J(q) - J_{\perp}/4} \right). \quad (3.48)$$

Taking into account  $P_5$  by replacing couplings  $J_{\alpha}$  with  $K_{\alpha}$  and the Greens function  $G_{\alpha\alpha}^J$  with

$$G_{\alpha\alpha}^K(\mathbf{q}) = \frac{1}{\omega - \frac{1}{2} [\epsilon_{\alpha}(\mathbf{q}) - \epsilon_{\alpha}(\bar{\mathbf{q}})] + \frac{1}{2} K_{\alpha}(\mathbf{q}) + W_{\alpha}(\mathbf{q})}, \quad (3.49)$$

we have the total renormalisation

$$\begin{aligned} \frac{\Delta J_{\perp}}{\Delta D} = & \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) \left[ J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}}) \sum_{\alpha} \frac{1}{1/G_{\alpha\alpha}^J(\mathbf{q}) - J_{\perp}/4} \right. \\ & \left. + K_f(\bar{\mathbf{q}}, \mathbf{q}) K_d(\mathbf{q}, \bar{\mathbf{q}}) \sum_{\alpha} \frac{1}{1/G_{\alpha\alpha}^K(\mathbf{q}) - J_{\perp}/4} \right]. \end{aligned} \quad (3.50)$$

### 1.3.5 Complete RG Equations

**Inter-layer Kondo coupling  $J_\perp$**

$$\begin{aligned}
\frac{\Delta J_\perp}{\Delta D} = & -\frac{1}{4} \int_{UV} d\mathbf{q} \rho(q) \int_{UV+IR} d\mathbf{k} \rho(k) \sum_\alpha \left[ J_\alpha^2(q, k) \left( \frac{1}{1/G_\alpha^J + 3J_\perp/4} + \frac{1}{1/G_\alpha^J - J_\perp/4} \right) \right. \\
& K_\alpha^2(q, k) \left( \frac{1}{1/G_\alpha^K + 3J_\perp/4} + \frac{1}{1/G_\alpha^K - J_\perp/4} \right) \left. \right] \\
& + \int_{UV} d\mathbf{q} \rho(q) \int_{UV+IR} d\mathbf{k} \rho(k) \sum_\alpha \frac{J_\alpha(q, k) K_\alpha(k, q)}{1/G_\alpha^{JK} - J_\perp/4} \\
& + \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) \sum_\alpha \left[ \frac{J_\alpha(\bar{\mathbf{q}}, \mathbf{q}) J_\alpha(\mathbf{q}, \bar{\mathbf{q}})}{1/G_{\alpha\alpha}^J(\mathbf{q}) - J_\perp/4} + \sum_\alpha \frac{K_\alpha(\bar{\mathbf{q}}, \mathbf{q}) K_\alpha(\mathbf{q}, \bar{\mathbf{q}})}{1/G_{\alpha\alpha}^K(\mathbf{q}) - J_\perp/4} \right]
\end{aligned} \tag{3.51}$$

**Direct Kondo coupling  $J_\alpha$**

$$\begin{aligned}
\frac{\Delta J_\alpha(k_1, k_2)}{\Delta D} = & - \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) + 4J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \\
& \left( \frac{1/4}{1/G_\alpha^J(\mathbf{q}) + 3J/4} + \frac{3/4}{1/G_\alpha^J(\mathbf{q}) - J/4} \right) \\
& + \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [K_\alpha(\mathbf{k}_1, \mathbf{q}) K_\alpha(\mathbf{q}, \mathbf{k}_2) - 4K_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \\
& \left( \frac{1}{1/G_\alpha^K + 3J/4} + \frac{1}{1/G_\alpha^K - J/4} \right) .
\end{aligned} \tag{3.52}$$

**Indirect Kondo coupling  $K_\alpha$**

$$\begin{aligned}
\frac{\Delta K_\alpha(k_1, k_2)}{\Delta D} = & - \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [K_\alpha(\mathbf{k}_1, \mathbf{q}) K_\alpha(\mathbf{q}, \mathbf{k}_2) + 4K_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \\
& \left( \frac{1}{4} \frac{1}{1/G_\alpha^K(\mathbf{q}) + 3J/4} + \frac{3}{4} \frac{1}{1/G_\alpha^K(\mathbf{q}) - J/4} \right) \\
& + \frac{1}{2} \int_{UV} d\mathbf{q} \rho(\mathbf{q}) [J_\alpha(\mathbf{k}_1, \mathbf{q}) J_\alpha(\mathbf{q}, \mathbf{k}_2) - 4J_\alpha(\mathbf{q}, \bar{\mathbf{q}}) W_\alpha(\mathbf{k}_1, \mathbf{q}, \bar{\mathbf{q}}, \mathbf{k}_2)] \\
& \left( \frac{1}{1/G_\alpha^J + 3J/4} + \frac{1}{1/G_\alpha^J - J/4} \right) .
\end{aligned} \tag{3.53}$$

## Propagators

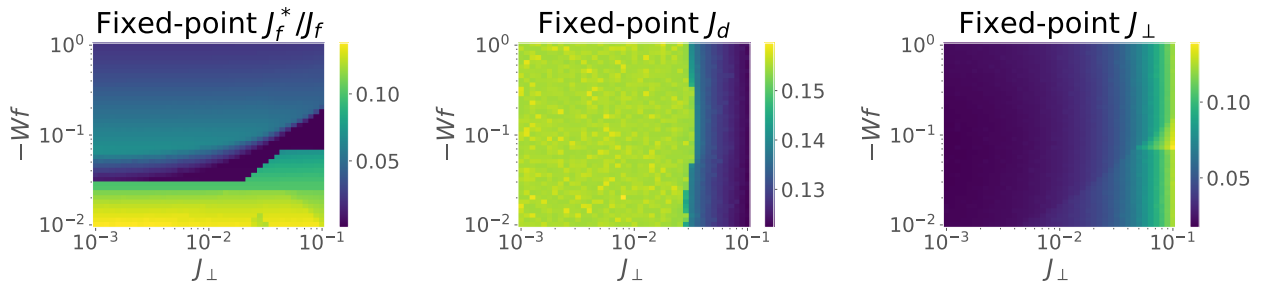
$$\begin{aligned}
 \text{Intra-layer spin-flip excitation: } G_{\alpha}^J(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_{\alpha}(\mathbf{q})|/2 + J_{\alpha}(\mathbf{q})/4 + W_{\alpha}(\mathbf{q})/2}, \\
 \text{Inter-layer spin-flip excitation: } G_{\alpha}^K(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_{\alpha}(\mathbf{q})|/2 + K_{\bar{\alpha}}(\mathbf{q})/4 + W_{\alpha}(\mathbf{q})/2}, \\
 \text{Both spins, one of the baths: } G^{JK}(\mathbf{q}) &= \frac{1}{\omega - |\epsilon_{\alpha}(\mathbf{q})|/2 + J_{\alpha}(\mathbf{q})/4 + K_{\bar{\alpha}}(\mathbf{q})/4 + W_{\alpha}(\mathbf{q})/2}, \\
 \text{Intra-layer spin-flip, two UV excitation: } G_{\alpha\alpha}^J(\mathbf{q}) &= \frac{1}{\omega - \frac{1}{2} [\epsilon_{\alpha}(\mathbf{q}) - \epsilon_{\alpha}(\bar{\mathbf{q}})] + \frac{1}{2} J_{\alpha}(\mathbf{q}) + W_{\alpha}(\mathbf{q})}, \\
 \text{Inter-layer spin-flip, two UV excitation: } G_{\alpha\alpha}^K(\mathbf{q}) &= \frac{1}{\omega - \frac{1}{2} [\epsilon_{\alpha}(\mathbf{q}) - \epsilon_{\alpha}(\bar{\mathbf{q}})] + \frac{1}{2} K_{\bar{\alpha}}(\mathbf{q}) + W_{\alpha}(\mathbf{q})}.
 \end{aligned} \tag{3.54}$$

## 1.4 Analysis for the simplified case of infinite dimensions

We first investigate the properties of the impurity model in the limit of infinite dimensions, on a Bethe lattice. This limit suppresses any momentum-space differentiation arising from low-dimensional lattices, and allows the study of real-space local properties.

We simulate the RG equations by adding small *negative* seed values of  $K_f$  and  $K_d$ . The other couplings are set as follows, in units of the conduction electron bandwidth:  $J_f = J_d = 0.1$ ,  $W_d = 0.0$ . The results are shown in Fig. 1.1, in the form of the renormalised low-energy fixed point values of the Kondo couplings  $J_f$  and  $J_d$  and the inter-layer coupling  $J_{\perp}$ , all as functions of  $W_f$  and  $J_{\perp}$ . We find that the  $d$ -layer Kondo coupling  $J_d$  (central panel) undergoes a suppression at larger values of the inter-layer coupling  $J_{\perp}$ . This coincides with an increase in the renormalised value of  $J_{\perp}$  (right panel) itself, indicating a transfer of correlations from within the  $d$ -layer to between the  $d$ - and  $f$ -layers. The couplings  $J_d$  is, however, largely unaffected by the bath correlation of the  $f$ -layer, as seen from the lack of any dependence of the boundary on  $W_f$ .

The  $f$ -layer coupling  $J_f$  (left panel) is suppressed with increasing  $|W_f|$  as well as increasing  $J_{\perp}$ . The phase space for  $J_f$  therefore splits, roughly, into four regions. The lower left region involves a dominant  $J_f$  coupling and displays local Fermi liquid excitations within the  $f$ -layer. The upper



**Figure 1.1:** Fixed-point values for the direct Kondo couplings  $J_{\alpha}$  and  $J_{\perp}$ .

left region has the in-plane correlation  $W_f$  driving  $J_f$  to irrelevance and leads to a Mott insulator within the layer, which can order magnetically if spontaneous symmetry breaking is allowed. The lower right region has the influence of both  $J_f$  and  $J_\perp$  and therefore involves the hybridisation of the local Fermi liquids in the two layers. Finally, the upper right region has the Mott state of the  $f$ -layer hybridising with the  $d$ -layer through  $J_\perp$ .

## 1.5 Infinite dimensions: Real-space correlation and phase diagram

In order to characterise the various phases, we compute a set of real-space correlations involving the impurity sites and the conduction bath sites hybridising with them. For small  $|W_f|$ , the  $f$ -layer is highly correlated but still itinerant due to the presence of local Fermi liquid excitations. In order to capture the  $J_\perp$ -driven transition in this regime, we track the inter-impurity compensation factor  $\langle \mathbf{S}_f \cdot \mathbf{S}_d \rangle$  that quantifies the strength of the singlet state between these two sites. As seen from the left panel in Fig. 1.2, the inter-impurity correlation (red curve) grows to large values as  $J_\perp$  is increased and eventually saturates close to the maximum value. This marks a transition from a phase in which the two metallic layers are decoupled and participate in transport independently, to one in which the local Fermi liquids hybridise and form local bound states.

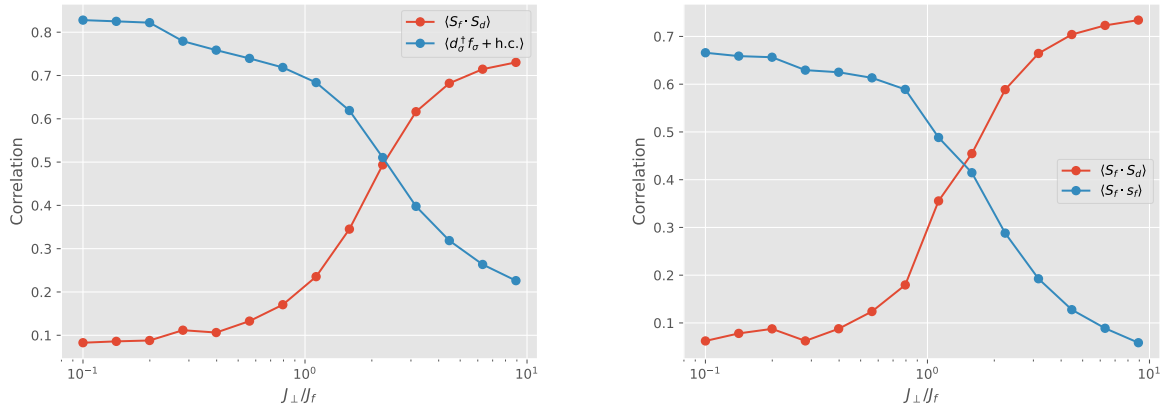


Figure 1.2

A complimentary picture is obtained in the same regime by studying the one-particle hybridisation between the  $f$ -impurity and its nearest-neighbour site  $f_0$ . While this measure is large for small  $J_\perp$ , indicating the presence of in-plane Kondo screening, and it gets suppressed as  $J_\perp$  is increased, revealing that the  $f$ -impurity is no longer hybridising with its own bath.

For large  $|W_f|$ , the  $f$ -layer is sufficiently strongly correlated to form a Mott insulator where the charge excitations are gapped, and the low-energy theory is described by a Heisenberg model. Here, we retain the inter-layer compensation factor as a measure but we replace the one-particle hybridisation with the spin-spin correlation  $\langle \mathbf{S}_f \cdot \mathbf{s}_{f0} \rangle$  between the  $f$ -spin and its nearest spin;

this change is necessitated by the charge gap in the spectrum. We again find that the inter-layer correlation grows as  $J_{\perp}$  is increased, while the intra-layer spin correlation shrinks. This marks a transition between a phase with no Kondo screening between the layers (and where the  $f$ -layer is potentially magnetically ordered) to one where the  $f$ -spin is Kondo screened by the other layer. In the context of heavy fermions, this marks a Kondo breakdown QCP between a magnetically-ordered phase with no (static) Kondo screening versus a heavy Fermi liquid phase where the  $f$ -spin participates in transport through its hybridisation with the other layer.

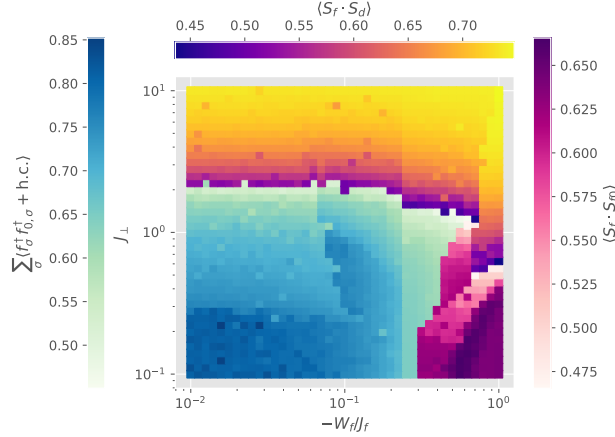


Figure 1.3

These conclusions are crystallised in the phase diagram in Fig. 1.3, computed using the three measures described above. For small  $J_{\perp}$  and  $|W_f|$  (lower left region), the two layers are metallic and mostly decoupled. Increasing  $J_{\perp}$  along the  $y$ -axis leads to a heavy Fermi liquid phase, while increasing  $|W_f|$  along the  $x$ -axis leads to the  $f$ -layer becoming insulating and decoupled from the other layer.

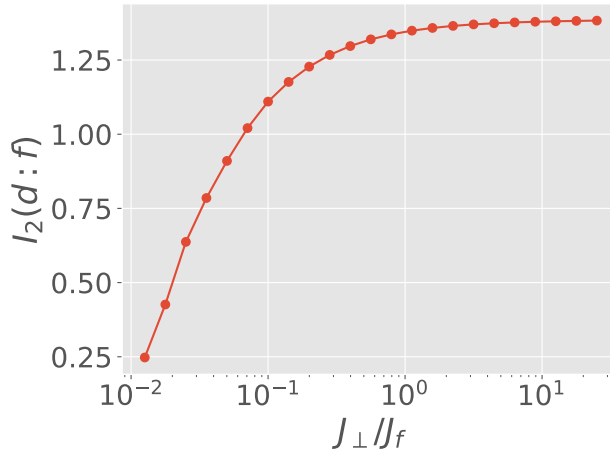


Figure 1.4



We have also characterised the transition using entanglement measures. Specifically, the mutual information (Fig. 1.4 between the  $f$ - and  $d$ -spins grows during the emergence of the heavy Fermi liquid, as  $J_\perp$  is increased. While this holds for both large and small  $|W_f|$ , there's one difference. Upon lowering  $J_\perp$ , the  $I_2$  for the case of small  $|W_f|$  vanishes much more rapidly compared to the case of large  $|W_f|$ . We believe the former is due to the gapless excitations present within the  $f$ -layer at small  $|W_f|$  that promote strong entanglement within the layer and hence compete with the inter-layer entanglement.

## 1.6 Infinite dimensions: Local spectral function

To further investigate the phases, we compute the local spectral functions of the  $f$ - and  $d$ -impurities. For weak  $J_\perp$ , the  $d$ -impurity always exhibits a central Kondo resonance, while the  $f$ -layer supports it only for weak  $W_f$ . For larger  $J_\perp$ , this resonance splits into two bands, in both the layers. These bands correspond to the two hybridised levels formed locally due to the mixing of the  $f$ - and  $d$ -impurities. For weak  $|W_f|$ , this is essentially the mixing of the two local Fermi liquids, while for large  $|W_f|$ , it arises purely from the LFL of the  $d$ -layer. At half-filling, the splitting of the central resonance into bands leads to a gap at zero frequency, and we get a Kondo insulator.

Away from half-filling, if the chemical potential lies in one of the bands, the systems becomes a heavy Fermi liquid. The value of  $W_f$  determines the amount of correlation in the heavy Fermi liquid. Larger values of  $|W_f|$  lead to narrow bands for the metal.

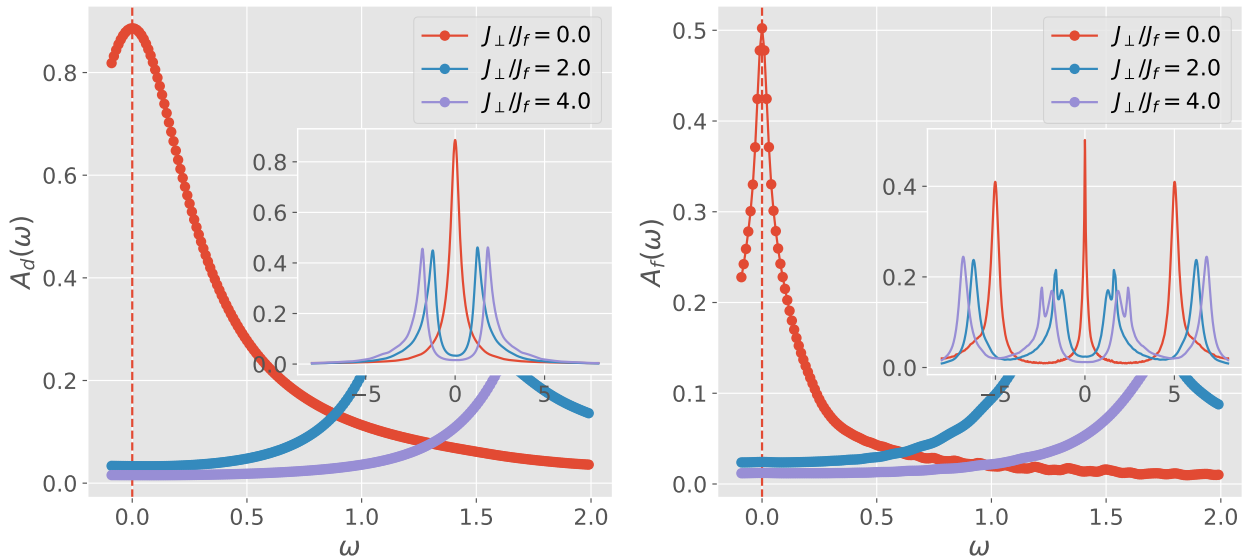


Figure 1.5

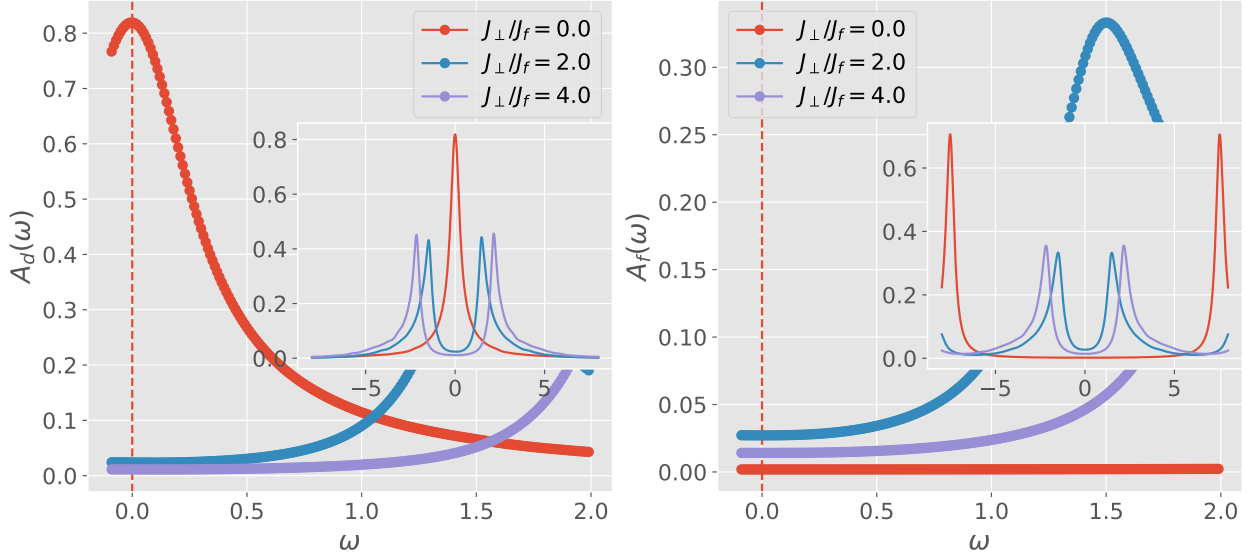


Figure 1.6

## 1.7 Numerical Implementation

Since the bath interaction doesn't renormalise, its functional form remains unchanged and can be substituted into the RG equations. In fact, we can write the expressions in a compact matrix form for efficient computation:

$$\frac{\Delta J_\alpha}{\Delta D} = -J_\alpha \mathcal{G}_\alpha J_\alpha^\dagger - 4\mathcal{W}_\alpha \text{Tr} [\mathcal{G}'_\alpha] , \quad (7.1)$$

where all objects are matrices.  $J_\alpha$  is the Kondo coupling matrix.  $\mathcal{G}_\alpha$  is a diagonal matrix with matrix elements

$$\mathcal{G}_\alpha[q, q] = \begin{cases} \rho(q) \left( \frac{1}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 + 3J/4} + \frac{3}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) + \mu/2 - J/4} \right), & \mathbf{q} \in \text{PS} , \\ \rho(q) \left( \frac{1}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 + 3J/4} + \frac{3}{8} \frac{1}{1/G_\alpha^{(0)}(\mathbf{q}) - \mu/2 - J/4} \right), & \mathbf{q} \in \text{HS} , \end{cases} \quad (7.2)$$

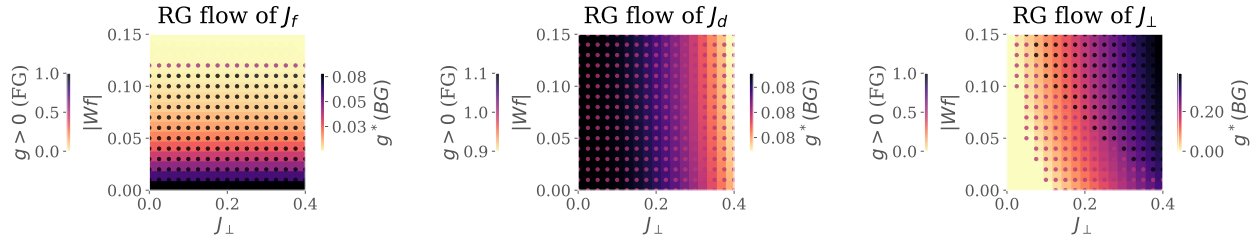
if  $q$  is in the UV subspace and zero otherwise.  $\mathcal{W}_\alpha$  is the bath interaction matrix whose matrix elements are given by  $\mathcal{W}_\alpha[k, k'] = \frac{W}{2} [\cos(k_x - k'_x) + \cos(k_y - k'_y)]$ . Finally,  $\mathcal{G}'_\alpha$  is another diagonal matrix whose matrix elements are defined as  $\mathcal{G}'_\alpha[q, q] = \mathcal{G}[q, q] J_\alpha[q, \bar{q}]$ .

The RG equation for the inter-layer coupling  $J$  can be written as

$$\frac{\Delta J}{\Delta D} = \frac{1}{2} \mathbf{J} \cdot \mathbf{G} , \quad (7.3)$$

where  $\mathbf{J}$  and  $\mathbf{G}$  are vectors defined by the elements  $\mathbf{J}[q] = J_f(\bar{\mathbf{q}}, \mathbf{q}) J_d(\mathbf{q}, \bar{\mathbf{q}})$  and

$$\mathbf{G}[q] = \rho(\mathbf{q}) \left[ \frac{1}{\omega - \frac{1}{2} \bar{\epsilon}_f(\mathbf{q}) + \frac{1}{2} J_f(\mathbf{q}) + W_f(\mathbf{q}) - \frac{1}{4} J} + \frac{1}{\omega - \frac{1}{2} \bar{\epsilon}_d(\mathbf{q}) + \frac{1}{2} J_d(\mathbf{q}) + W_d(\mathbf{q}) - \frac{1}{4} J} \right] . \quad (7.4)$$



**Figure 1.7:** RG phase diagrams for various Kondo couplings:  $f$ -layer (left),  $d$ -layer (middle) and inter-layer (right). For each plot, the background color (quantified by the right colorbar) tracks the fixed point value of the coupling as a function of the inter-layer couplings  $J_{\perp}$  and the  $f$ -layer bath interaction  $|W_f|$ . The marker color complements this by tracking whether the coupling is relevant, intermediate-coupling or irrelevant in the RG sense, by taking the values 1, 1/2 or 0 respectively.

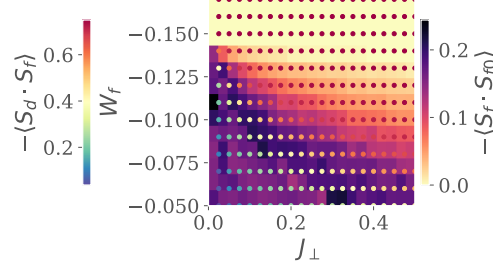
## 1.8 Renormalisation Group Flows of Kondo Couplings: RG Phase Diagram

We first analyse the model by obtaining unitary RG flows of three kinds of Kondo couplings. Fig. 1.7 shows the fixed point values of the couplings under renormalisation to a low-energy theory. We view these values as a function of the  $f$ -layer bath interaction  $W_f$  and inter-layer Kondo coupling  $J_{\perp}$ . In each panel, the right (left) colorbar quantifies the background (marker) color of the plot. The background displays the fixed point value of each coupling whereas the marker color tracks whether the couplings is relevant (1), intermediate coupling (0.5) or irrelevant (0).

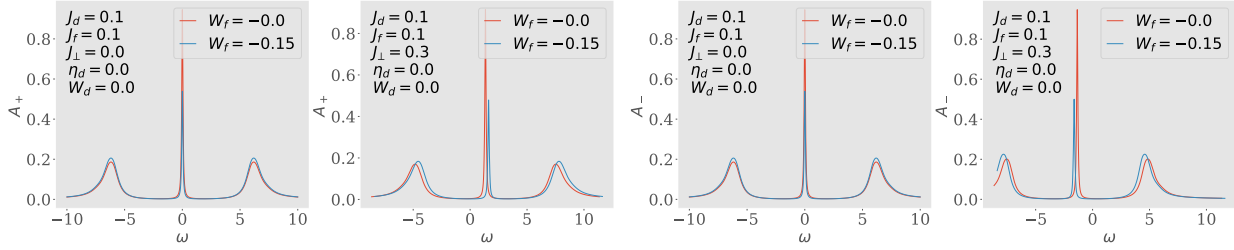
The Kondo coupling  $J_d$  (middle panel) of the  $d$ -layer remains relevant for the entirety of the chosen parameter regime. The Kondo coupling  $J_f$  (left panel) of the  $f$ -layer becomes RG-irrelevant for larger values of  $|W_f|$ ; this leads to an in-plane Mott transition of the kind observed in our earlier work, where the local Fermi liquid excitations of the  $f$ -impurity give way to a pseudogapped non-Fermi liquid Mott metal phase which ultimately destabilises into a local moment phase. The RG flow of  $J_f$  does not appear to be significantly affected by the variation of  $J_{\perp}$ . Concomitant with the irrelevance of  $J_f$  under increasing  $|W_f|$ , the inter-plane coupling  $J_{\perp}$  (right panel) becomes RG-relevant, as seen from the darkening markers as we go to higher  $|W_f|$ .

These simulations motivate an RG-based phase diagram with the following regimes: (i) weak  $J_{\perp}$ , weak  $|W_f|$ : decoupled metallic layers, each layer participating in transport independently, (ii) strong  $J_{\perp}$ , strong  $|W_f|$ : Kondo insulator regime, where the local moments of the  $f$ -layer hybridise with the itinerant electrons of the  $d$ -layer to form a band insulator with the chemical potential situated in the hybridisation gap, and (iii) weak  $J_{\perp}$ , strong  $|W_f|$ : the so-called RKKY regime, where the  $f$ -layer forms local moments which decouple from the  $d$ -layer and are then free to order magnetically, and (iv) strong  $J_{\perp}$ , weak  $|W_f|$ : this is similar to the Kondo insulator regime, except here the  $f$ -electrons remain itinerant and it is they who hybridise with the  $d$ -electrons to again form two bands separated by a hybridisation gap.

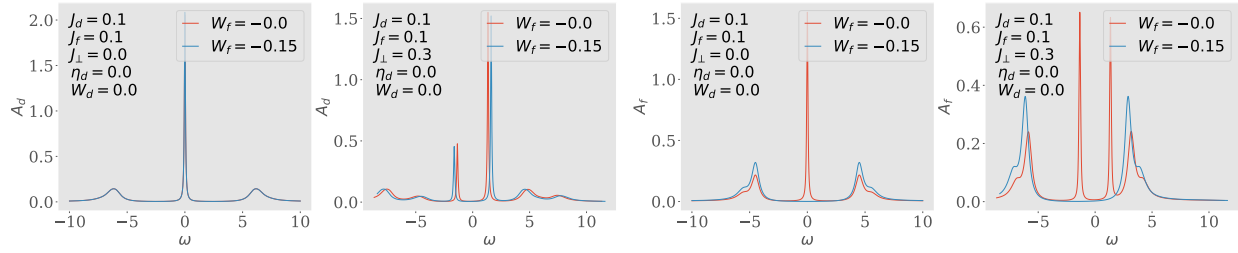
## 1.9 Results at half-filling



**Figure 1.8:** Left: Real-space spin correlations for the auxiliary model. Background colour (right colourbar) represents that between the  $f$ -impurity and its nearest-neighbour sites, while the marker colour (left colourbar) represents that between the  $f$ -impurity and the  $d$ -impurity. While the former shrinks as  $J_{\perp}$  or  $|W_f|$  is increased, the latter grows. This marks a transfer of entanglement from intra-plane channel into inter-plane channel. Right:



**Figure 1.9:** Spectral functions of the tiled model, for excitations in the bonding (two on the left) and antibonding (two on the right) basis, defined by the Greens function  $G_{k,\pm} = G(k_d \pm k_f; \omega)$ . There is no hybridisation for  $J_{\perp} = 0$  (first and third plots), and the only effect of  $W_f$  is to shrink the height of the central resonance. Introducing a non-zero  $J_{\perp}$  splits the central resonance into an upper band (second from left) and a lower band (first from right). This leads to a Kondo insulator, where  $\omega = 0$  resides inside the hybridisation gap.



**Figure 1.10:** Spectral functions  $A_d$  and  $A_f$  of the auxiliary model, for excitations in the  $d$ - and  $f$ -layers respectively. Both show a central Kondo resonance for  $J_\perp = 0$  which splits into two bands upon adding a non-zero  $J_\perp$ . Increasing  $|W_f|$  to large values gaps out the local Fermi liquid excitations of  $A_f$ .

# Bibliography

- [1] Abhirup Mukherjee, N S Vidhyadhiraja, A Taraphder, and Siddhartha Lal. Kondo frustration via charge fluctuations: a route to mott localisation. *New Journal of Physics*, 25(11):113011, nov 2023.